

MORITZ GRAUER

Karlsruhe, Germany
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Rendering engineer with a passion for games. Specialized in ray tracing optimization, cache-efficient algorithms, and real-time graphics systems.

TECHNICAL SKILLS

Graphics Programming

APIs: Vulkan, OpenGL

Rendering: Ray/Path Tracing, Monte Carlo Methods, MLT, Photon Mapping, Real-time & Offline

Programming

C, C++, Python, Java, C#, CUDA, GLSL

Tools & Engines

Unity, Unreal Engine, Blender, Git, ImGui, GLFW

EXPERIENCE

Computer Graphics Researcher

Feb 2022 – Apr 2026

Karlsruhe Institute of Technology (KIT)

- Published research on ray tracing memory optimization
- Developed custom rendering engine with ray/path tracing support
- Supervised Bachelor/Master theses and organized Computer Graphics lectures/exercises

QA Engineer

Oct 2020 – Sep 2021

Gameforge (Internship + Working Student)

- Tested gameplay functionality and quest systems across multiple MMO titles
- Debugged and verified in-game rewards, progression systems, and content updates
- Collaborated with dev teams on quality standards and bug resolution

Programming Tutor

Oct 2016 – Jul 2019

Karlsruhe Institute of Technology (KIT)

- Mentored CS students in programming and algorithms

PROJECTS

Global Game Jam 2026 – Quack 3

(globalgamejam.org/games/2026/quack-3-5)

Playable FPS merging 2D art with 3D gameplay. Built in Unity without third-party assets. Handcrafted art, SFX and shaders.

Custom Graphics Engine – VexEngine

(github.com/Vexury/VexEngine)

A C++ graphics engine built from scratch — path tracing, Cook-Torrance GGX, dual OpenGL/Vulkan backend, BVH acceleration, full editor UI and more.

EDUCATION

M.Sc. Informatics

2018 – 2021

Karlsruhe Institute of Technology (KIT)

Specialization: Computer Graphics & AI

Exchange Semester

2018 – 2019

Chalmers University, Gothenburg, Sweden

B.Sc. Informatics

2015 – 2018

Karlsruhe Institute of Technology (KIT)

PUBLICATIONS

Minimizing Ray Tracing Memory Traffic through Quantized Structures and Ray Stream Tracing

arXiv, 2023

Novel approach to reducing memory bandwidth in ray tracing through quantized BVH structures

HONORS

Studienstiftung des deutschen Volkes

German Academic Scholarship Foundation (2018 – 2021)

LANGUAGES

German: Native

English: Fluent (written & spoken)

French, Spanish, Swedish: Basic